

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8/2	(a)	(i)		A pupil B Iris both for one mark	1			1		
		(ii)		In brighter light (1) {Muscles/ iris} (contract to) reduce pupil {diameter/size} (1) Restricts light entering the eye, so retina is protected (from damage) (1)	1	2		3		
		(iii)		Fast and {involuntary/ automatic} Reject reference to protection	1			1		
	(b)	(i)		Reduced reaction time/ faster reactions		1		1		
		(ii)		66.7 = 2 marks Incorrect answer but correct method = 1 mark $(25 - 15)/15 \times 100 = 66.7(\%) = 1 \text{ mark}$ $67/ 66.666^\circ \% = 1 \text{ mark}$		2		2	2	
		(iii)		Any two (x1) from: - the older the person {the fewer notes hit in 0.2 seconds/ the slower the reactions /the increased reaction time/ ORA - after {practice/repeats} {age 50/ older people} {improvement doubles/ improve more}, (this is more than for younger ages)		1	1	2		
		(iv)	I	Any two (x1) from Equal numbers male/female OR one gender only Same experience of keyboard playing Handedness Eyesight No alcohol/ caffeine			2	2		3
			II	Larger sample size/ wider range of ages/ more people of different ages (1)			1	1		
				Question 8 total	3	6	4	13	2	3